Decentralized Optimization Lower/Upper Bound Table

1 Notation

Objective functions, accuracy, and stochastic model. F denotes global objective, $F(x) = N^{-1}\sum_{i=1}^{N} F_i(x)$; F_i denotes local objective at node i; $F^\star$ denotes optimal value, $F^\star = \inf_x F(x)$; ϵ denotes target accuracy, $\epsilon > 0$; $\mathbb{E}$ denotes expectation operator; σ^2 denotes stochastic-gradient variance bound, $\mathbb{E}\|g_i(x;\xi_i) - \nabla F_i(x)\|^2 \le \sigma^2$; $g_i(x;\xi_i)$ denotes the stochastic first-order oracle output at node i, evaluated at x using local random sample ξ_i; σ denotes standard-deviation bound, $\sigma = \sqrt{\sigma^2}$; ζ^2_{stat} denotes uniform cross-node gradient-heterogeneity bound satisfying $N^{-1}\sum_{i=1}^{N}\|\nabla F_i(x) - \nabla F(x)\|^2 \le \zeta^2_{\text{stat}}$ for all x [48]; δ_{ht} denotes failure probability in a high-probability guarantee, $\delta_{\text{ht}} \in (0,1)$ [10] [26] [52]; Pr denotes probability under the stochastic oracle and algorithm randomness [26] [52].

Problem geometry and parameters. Δ denotes initial optimality gap, $\Delta = F(x^0) - F^\star$; Δ_0 denotes initial objective-gap bound, $\Delta_0 \ge F(x^0) - F^\star$ [27] [41] [11]; N denotes number of nodes; T denotes iteration horizon; T_{out} denotes outer-iteration horizon when each outer step contains a local batch or multiple gossip rounds; K_{loc} denotes number of local component functions, where f_{ij} is component j at node i and $F_i(x) = K_{\text{loc}}^{-1}\sum_{j=1}^{K_{\text{loc}}} f_{ij}(x)$; S denotes epoch index [32]; L denotes smoothness constant satisfying $\|\nabla F_i(x) - \nabla F_i(y)\| \le L\|x-y\|$ for every node i; $\bar{L}$ denotes mean-squared component smoothness constant satisfying $(NK_{\text{loc}})^{-1}\sum_{i=1}^{N}\sum_{j=1}^{K_{\text{loc}}}\|\nabla f_{ij}(x) - \nabla f_{ij}(y)\|^2 \le \bar{L}^2\|x-y\|^2$ [9]; μ denotes strong-convexity constant satisfying $h(y) \ge h(x) + \langle g, y-x\rangle + \frac{\mu}{2}\|y-x\|^2$ for $g \in \partial h(x)$; κ denotes condition number, $\kappa = L/\mu$; κ_s denotes mean-squared finite-sum condition number, $\kappa_s = \bar{L}/\mu$ [9] [22]; $\kappa_{s,\text{VR}}$ denotes VR-EXTRA/DIGing stochastic condition number, $\kappa_{s,\text{VR}} = \mu^{-1}\max_i K_{\text{loc}}^{-1}\sum_{j=1}^{K_{\text{loc}}} L_{ij}$, where L_{ij} is the smoothness constant of component j at node i [20]; q_{DPS} denotes DPSVRG exponent $q_{\text{DPS}} = \log(1/r_{\text{DPS}})/\log\beta_{\text{DPS}} \in (0,1)$ [32]; M denotes uniform local subgradient bound satisfying $\|g_i(x)\| \le M$ for every $g_i(x) \in \partial F_i(x)$; R denotes initial-distance bound satisfying $\|x^0 - x^\star\| \le R$; c denotes initial error constant in the primal-dual PL bound [41]; α_{DSP} denotes positive constant [46]; ℓ_{DSP} denotes D-SPIDER-SFO complexity coefficient [46]; C_{DDA} denotes DDA constant coefficient [38]; C_2 denotes DAGP heterogeneity coefficient [27]; $\bar{\zeta}$ denotes root-mean-square local-gradient heterogeneity at a global optimum, $\bar{\zeta} = (N^{-1}\sum_{i=1}^{N}\|\nabla F_i(x^\star)\|^2)^{1/2}$ [44]; p denotes higher-order smoothness degree, an integer $p \ge 2$ such that $\|\mathcal{D}^p F_i(X) - \mathcal{D}^p F_i(Y)\|_* \le L_{p,*}\|X - Y\|$ for some $L_{p,*} > 0$, where $\mathcal{D}^p$ is the p-th derivative [39]; Ψ^0 denotes initial MG-Skip Lyapunov value [23]; C_0 denotes DAGP initialization coefficient; κ_{snc} denotes sum-of-nonconvex composite condition number, $\kappa_{\text{snc}} = (L_{\text{snc}} + \sqrt{\ell_1\ell_2})/\mu$, where L_{snc} is the smooth aggregate's Lipschitz-gradient constant (the source's L), μ is the total strong convexity, and component Bregman remainders lie between $-\ell_2\|x-y\|^2/2$ and $\ell_1\|x-y\|^2/2$, with $\ell_2 \ge \ell_1 > 0$ [50]; $L_{\text{rms,loc}}$ denotes uniform node-local RMS component smoothness: for every node i, $K_{\text{loc}}^{-1}\sum_j\|\nabla f_{ij}(x) - \nabla f_{ij}(y)\|^2 \le L^2_{\text{rms,loc}}\|x-y\|^2$ [22]; L_F denotes smoothness constant of the global objective F, used when a source distinguishes global smoothness from node or component smoothness [9]; C_{DPS} denotes full DPSVRG initial-gap, distance, and estimator-error constant stated in the row-37 assumption note [32]; M_ϕ denotes uniform subgradient bound for the nonsmooth convex component ϕ_i in row 55 [45].

Topology, network, and constraints. κ_A denotes equilibrium skewness of row-stochastic matrix A, $\kappa_A = \max_i(\pi_A)_i/\min_i(\pi_A)_i$ [47]; κ_π denotes equilibrium skewness of column-stochastic matrix W_π, $\kappa_\pi = \max_i \pi_i/\min_i \pi_i$ [7]; ρ denotes spectral gap, $\rho = 1 - \|W - N^{-1}\mathbf{1}_N\mathbf{1}_N^\top\|_2$; ρ_{K} denotes reference-specific spectral gap, $\rho_{\text{K}} = 1 - \lambda_2(P)$, for the fixed symmetric doubly stochastic gossip matrix P, where $\lambda_2(P)$ is the second-largest eigenvalue of P [11]; β_{H} denotes one-step mixing factor satisfying $\|W^t - N^{-1}\mathbf{1}_N\mathbf{1}_N^\top\|_2 \le \beta_{\text{H}} < 1$ [8]; β_A denotes row-stochastic mixing factor of A, $\beta_A = \|A - \mathbf{1}_N\pi_A^\top\|_{\pi_A} \in [0,1)$, where $\Pi_A = \text{diag}(\pi_A)$ and $\|B\|_{\pi_A} := \|\Pi_A^{1/2}B\Pi_A^{-1/2}\|_2$ is the induced weighted matrix norm [47]; β_π denotes column-stochastic mixing factor of W_π, $\beta_\pi = \|W_\pi - \pi\mathbf{1}_N^\top\|_\pi \in [0,1)$, where $\Pi = \text{diag}(\pi)$ and $\|B\|_\pi := \|\Pi^{-1/2}B\Pi^{1/2}\|_2$ is the induced weighted matrix norm [7]; δ_{net} denotes normalized network contraction rate, $\delta_{\text{net}} := p_{\text{mix}}/\tau_{\text{mix}}$, where $1 - p_{\text{mix}}$ is the τ_{mix}-step expected squared-consensus-error factor [44]; W denotes doubly stochastic mixing matrix $W \in \mathbb{R}^{N\times N}$ satisfying $W\mathbf{1}_N = \mathbf{1}_N$ and $\mathbf{1}_N^\top W = \mathbf{1}_N^\top$; $\omega(\widehat{W})$ denotes smallest positive eigenvalue of the positive semidefinite gossip/Laplacian operator $\widehat{W}$ [12]; $C_{\mathbf{S}}$ denotes DAGP network-and-algorithm factor satisfying $C_{\mathbf{S}} > -\lambda_{\min}(\mathbf{S})$, where $\mathbf{S}$ is the theorem matrix determined by the gossip matrices and algorithm parameters [27]; W_π denotes primitive column-stochastic mixing matrix satisfying $\mathbf{1}_N^\top W_\pi = \mathbf{1}_N^\top$ [7]; π denotes normalized positive right Perron vector of W_π, satisfying $W_\pi\pi = \pi$ and $\mathbf{1}_N^\top\pi = 1$ [7]; π_A denotes normalized positive left Perron vector of row-stochastic A, satisfying $\pi_A^\top A = \pi_A^\top$ and $\pi_A^\top\mathbf{1}_N = 1$ [47]; p_{mix} denotes blockwise expected consensus-contraction amount in the definition of δ_{net} [44]; τ_{AC} denotes AC-GT mixing-window coefficient, $\tau_{\text{AC}} = \eta_{\text{AC}}\bar{\tau}d_{\mathcal{G}}$, where $\bar{\tau}$ is the joint-connectivity window, $d_{\mathcal{G}}$ is the graph diameter, and $\eta_{\text{AC}} \in \mathbb{N}$ satisfies the cited block-contraction condition [18].

Iterates and outputs. x_i denotes local iterate/output at node i; x_i^t is its value at iteration t, while the unindexed form is the theorem's returned output; $\bar{x}$ denotes network-average iterate/output, $\bar{x}^t = N^{-1}\sum_{i=1}^N x_i^t$; the unindexed $\bar{x}$ denotes the averaged output specified by the cited theorem; $x_{o,i}$ denotes output selected by node i from its internal memory [4]; x_o^T denotes network-average output, $x_o^T = N^{-1}\sum_{i=1}^N x_{a,i}^T$, where $x_{a,i}^T$ is node i's component of the averaged primal vector x_a^T [4]; $\bar{x}_{i,T}$ denotes time-averaged output at node i, $\bar{x}_{i,T} = T^{-1}\sum_{t=0}^{T-1} x_i^t$ [27]; $\bar{y}_T$ denotes averaged primal output after T iterations [29]; $\hat{x}_i(T)$ denotes time-averaged output at node i, $\hat{x}_i(T) = T^{-1}\sum_{t=1}^T x_i^t$ [30]; $\tilde{x}_i^T$ denotes time-averaged output at node i, $\tilde{x}_i^T = T^{-1}\sum_{t=1}^T x_i^t$ [31]; $\hat{x}$ denotes the algorithm output used to evaluate the accuracy criterion in the corresponding source; its specific construction follows the theorem cited in that row [9] [8] [6] [10] [21] [22] [26] [40] [49]; $\hat{y}$ denotes tuple of node-wise time-averaged outputs [33]; $\tilde{x}$ denotes virtual centralized output after epoch S in DPSVRG [32]; $\tilde{x}_i$ denotes weighted time-averaged output at node i, $\tilde{x}_i = A_T^{-1}\sum_{t=1}^T a_t x_i^t$, where $A_T = \sum_{t=1}^T a_t$ and $a_t > 0$ are theorem-specified averaging weights [25] [34]; $x^\star$ denotes optimizer of F, $x^\star \in \arg\min_x F(x)$; X denotes stacked local iterate/output, $X^t = [(x_1^t)^\top, \ldots, (x_N^t)^\top]^\top$; the unindexed form is the returned stacked output; $X^\star$ denotes stacked consensus optimizer, $X^\star = \mathbf{1}_N \otimes x^\star$; y denotes consensus auxiliary primal iterate in DDA, $y^t = \nabla d^*(-\bar{z}^t)$, where d^* is the conjugate of the prox function [38]; $\mathbf{y}$ denotes stacked auxiliary iterate in DDA, $\mathbf{y}^t = \mathbf{1}_N \otimes y^t$ [38]; z denotes node-wise dual accumulator z_i^t in DDA [38]; $\bar{z}$ denotes network-average auxiliary iterate, $\bar{z}^t = N^{-1}\sum_{i=1}^N z_i^t$ [38]; z_{VR}^k denotes VR-EXTRA/DIGing auxiliary primal iterate returned at iteration k [20]; $\bar{x}_u^k$ denotes left-Perron-weighted aggregate $\bar{x}_u^k = N^{-1}(u^\top \otimes I)X^k$, where $u^\top R = u^\top$ and $u^\top \mathbf{1}_N = N$ [36]; $\bar{x}_{\pi_{\mathrm{SAB}}}^k$ denotes S-AB Perron-weighted aggregate $\bar{x}_{\pi_{\mathrm{SAB}}}^k = (\pi_{\mathrm{SAB}}^\top \otimes I)X^k$, where π_{SAB} is the normalized positive left Perron vector of its row-stochastic mixing matrix [43]; $\bar{x}_{\pi_A}^t$ denotes row-stochastic Perron-weighted aggregate $\bar{x}_{\pi_A}^t = (\pi_A^\top \otimes I)X^t$ [47]; $\tilde{y}_{\mathrm{DA}}^T$ denotes dual-averaging auxiliary output: the time average of the source's prox points formed from averaged dual accumulators; generally not the mean of local primal outputs [25]; $\bar{y}_{\mathrm{PM}}^T$ denotes PMGT-KatyushaX network-average output after T epochs, $N^{-1}\sum_i y_i^T$ [50].

Targets and stationarity measures. δ_{G} denotes Goldstein-stationarity radius; G denotes Bregman distance of the stacked separable objective, $G(X, X^\star) = \sum_{i=1}^N [F_i(X_i) - F_i(x^\star) - \langle \nabla F_i(x^\star), X_i - x^\star\rangle]$, where X_i is node i's block of X [5]; $\mathbf{s}$ denotes proximal-gradient mapping of $F + h$, $\mathbf{s}(x) = \alpha^{-1}[x - \mathrm{prox}_{\alpha h}(x - \alpha\nabla F(x))]$, where h is the convex regularizer and $\alpha > 0$ [40]; G_a denotes DDA gradient mapping, $G_a(x, z) = a^{-1}[\nabla d^*(-z) - \nabla d^*(-z - \nabla f(x))]$ [38]; $\bar{g}$ denotes node-averaged local gradient at the stacked iterate, $\bar{g}(X) = N^{-1}\sum_{i=1}^N \nabla F_i(X_i)$ [39]; ∇F denotes gradient of the global objective; $\|\cdot\|$ denotes Euclidean norm of $v \in \mathbb{R}^d$, $\|v\| = (\sum_{r=1}^d v_r^2)^{1/2}$; $\|\nabla F(\hat{x})\|_{\delta_{\mathrm{G}}}$ denotes Goldstein stationarity measure with radius δ_{G}; $\|\cdot\|_*$ denotes nuclear norm of a matrix X, $\|X\|_* = \sum_r \sigma_r(X)$ [39]; $\|\cdot\|_{\mathbb{E}}$ denotes root-mean-square norm of a random vector v, $\|v\|_{\mathbb{E}} = (\mathbb{E}\|v\|^2)^{1/2}$ [38]; $\|\cdot\|_F$ denotes Frobenius norm of $X \in \mathbb{R}^{a\times b}$, $\|X\|_F = (\sum_{r=1}^a \sum_{s=1}^b X_{rs}^2)^{1/2}$; V^t denotes source-defined primal-dual Lyapunov value combining squared primal error, squared dual error, and gradient-estimator variance [12]; $\mathcal{E}_{\mathrm{obj}}$ denotes average local-objective error, $\mathcal{E}_{\mathrm{obj}}(y_1, \ldots, y_N) = N^{-1}\sum_{i=1}^N F_i(y_i) - F^\star$ [33].

Asymptotic notation. $O(\cdot)$ denotes asymptotic upper bound, $a = O(b)$ when $a \le Cb$ for a constant $C > 0$; $\Omega(\cdot)$ denotes asymptotic lower bound, $a = \Omega(b)$ when $a \ge cb$ for a constant $c > 0$; $\widetilde{O}(\cdot)$ denotes asymptotic upper bound hiding poly-logarithmic factors, $a = \widetilde{O}(b)$.

Indices and algorithmic parameters. i denotes node index; j denotes local component index; t denotes iteration index; η denotes positive stepsize [18] [36]; θ denotes linear contraction parameter $\theta \in (0, 1)$ in the factor $(1-\theta)^t$ [41]; η_{alg} denotes positive DAGP auxiliary parameter [27]; p_{comm} denotes communication probability, $p_{\mathrm{comm}} \in (0, 1]$ [23]; w_t denotes positive theorem weights used to average strongly-convex decentralized-SGD objective errors [44]; W_T denotes cumulative weight, $W_T = \sum_{t=0}^T w_t$ [44]; s denotes candidate iteration index in $\min_{1\le s\le t}$; α_{DAGP} denotes DAGP gradient stepsize, $\alpha_{\mathrm{DAGP}} > 0$.

Status labels. In the Status column, EXACT means that the LB and UB match in order; LOG means that they differ only by poly-logarithmic factors; COND means that matching requires a special metric or additional conditions; GAP means that the known LB and UB have different orders; LB? means that only an upper bound was retained; UB? means that only a lower bound was retained; and EMPTY means that zero exact decentralized LB/UB claim sides were retained within the recorded search scope.

2 Tags

Fixed scope. Every row concerns vanilla decentralized optimization with peer-to-peer gossip, synchronous communication rounds, and no communication compression. These four fixed attributes are therefore omitted from the row labels; the structured problem tags retain them for taxonomy and audit purposes.

Objective form. SUM denotes a deterministic distributed aggregate $F(x) = \frac{1}{N}\sum_{i=1}^N F_i(x)$, where each local objective F_i is deterministic. A SUM+SFO row is used when the deterministic local objective is accessed by a sampled oracle: either explicit finite-sum component sampling or an unbiased noisy-gradient oracle for F_i; oracle-only stochasticity is disclosed by a Remark marker. STO denotes an expectation objective $F(x) = \frac{1}{N}\sum_{i=1}^N \mathbb{E}_{\xi_i}[f_i(x; \xi_i)]$, where ξ_i is a local random sample and the objective is accessed through sampled stochastic first-order information. Thus SUM/STO describes the objective model, whereas the oracle tag below describes what one local query returns.

Geometry and regularity. SC means strong convexity with parameter $\mu > 0$; C means convexity without an assumed positive strong-convexity parameter; NC means that convexity is not assumed; and PL means the classical differentiable Polyak–Lojasiewicz condition $\frac{1}{2}\|\nabla F(x)\|^2 \geq \mu(F(x) - F^\star)$. The classification follows the level used by the cited statement (global objective, local objectives, or both), with stronger local assumptions retained as caveats. SM means differentiability with an L-Lipschitz gradient at the theorem's stated level; NS means that the total objective is not assumed differentiable with an L-Lipschitz gradient and the result instead uses nonsmooth first-order, subgradient, or proximal information. COMP marks a composite objective consisting of a smooth loss and an explicit nonsmooth/proximable regularizer; it supplements rather than replaces NS.

Local oracle. FO is an exact local first-order oracle (the exact gradient in the smooth case, or the exact first-order mapping specified by the paper); SFO is a sampled stochastic or finite-sum component oracle returning $g_i(x; \xi_i)$ under the cited theorem's bias, variance, and independence assumptions; and SUBG is a deterministic local subgradient oracle returning $g_i(x) \in \partial F_i(x)$. All displayed complexities count per-node local oracle calls, not total calls summed over the network. PROX means that the algorithm additionally assumes exact evaluation of the regularizer's proximal operator; these side-access calls are not included in the displayed FO/SFO counts.

Network evolution. STAT means that one communication graph and its mixing operator remain fixed over the run. TV means that the graph or mixing operator may vary by communication round, subject to the connectivity and mixing assumptions stated by the cited theorem.

3 Main Table

Table 1: Oracle-query complexity map. Tags describe the setting, Measure gives the target, LB/UB report oracle-query complexity only, Status marks exact matches or remaining gaps, and superscript markers link to Remark Notes.

No.	Problem Type	Measure	Lower Bound	Upper Bound	Status	Reference (LB)	Reference (UB)
1	SUM SC SM FO TV S12	$F(x_i) - F^\star \le \epsilon$ M3	$\Omega(\sqrt{\kappa}\log(1/\epsilon))$	$O(\sqrt{\kappa}\log(1/\epsilon))$ N2,N1	EXACT	[1], Theorem 3	[1], Theorem 6
2	SUM C NS SUBG STAT	$F(\bar{x}) - F^\star \le \epsilon$	$\Omega(M^2R^2/\epsilon^2)$ A10,A11	$O(M^2R^2/\epsilon^2)$ A10	EXACT	[2], Theorem 3	[2], Theorem 5
3	SUM SC SM FO STAT S16	$F(x_i) - F^\star \le \epsilon, \forall i$	$\Omega(\sqrt{\kappa}\log(1/\epsilon))$ O3	$O(\sqrt{\kappa}\log(1/\epsilon))$ O3	COND	[3], Theorem 2	[3], Theorem 4
4	SUM SC NS SUBG TV S12,S13	$F(x_{o,i}) - F^\star \le \epsilon_{\text{LB}}$; $F(x_o^T) - F^\star \le \epsilon_{\text{UB}}$ M4	$\Omega(M^2/(\mu\epsilon))$	$O(M^2/(\mu\epsilon))$	COND	[4], Theorem 1	[4], Theorem 3
5	SUM C SM FO STAT S12	$G(X, X^\star) \le \epsilon$ M1	$\Omega(\sqrt{LR^2/\epsilon} + R\sqrt{N}\,\bar{\zeta}/\epsilon)$	$O(\sqrt{LR^2/\epsilon} + R\sqrt{N}\,\bar{\zeta}/\epsilon)$ A9	COND	[5], Theorem 2; Corollary 3	[5], Theorem 7
6	SUM C NS SUBG TV S12,S13	$F(x_{o,i}) - F^\star \le \epsilon_{\text{LB}}$; $F(x_o^T) - F^\star \le \epsilon_{\text{UB}}$ M4	$\Omega(M^2R^2/\epsilon^2)$	$O(M^2R^2/\epsilon^2)$	COND	[4], Theorem 2	[4], Theorem 4
7	STO NC SM SFO STAT S3	$\mathbb{E}\|\nabla F(\hat{x})\|^2 \le \epsilon_{\text{LB}}$; $\frac{1}{T_{\text{out}}}\sum_{t=1}^{T_{\text{out}}}\mathbb{E}\|\nabla F(\bar{x}^{(t)})\|^2 \le \epsilon_{\text{UB}}$ M5	$\Omega(\Delta L\sigma^2/(N\epsilon^2) + \Delta L/(\sqrt{\rho}\epsilon))$	$\widetilde{O}(\Delta L\sigma^2/(N\epsilon^2) + \Delta L/(\sqrt{\rho}\epsilon))$ A3	COND	[6], Table 6	[6], Theorem 4
8	STO NC SM SFO STAT S11,S14	$\frac{1}{T_{\text{out}}+1}\sum_{t=0}^{T_{\text{out}}}\mathbb{E}\|\nabla F(\bar{x}^{(t)})\|^2 \le \epsilon$	$\Omega(\sigma^2 L\Delta/(N\epsilon^2) + (1+\log\kappa_\pi)L\Delta/((1-\beta_\pi)\epsilon))$ G7	$\widetilde{O}(\sigma^2 L\Delta/(N\epsilon^2) + (1+\log\kappa_\pi)L\Delta/((1-\beta_\pi)\epsilon))$ N3	COND	[7], Theorem 1; Equation (11)	[7], Equation (14)
9	STO NC SM SFO TV S2,S4,S13	$\mathbb{E}\|\nabla F(\hat{x})\|^2 \le \epsilon_{\text{LB}}$; $\frac{1}{T+1}\sum_{k=0}^{T}\mathbb{E}\|\nabla F(\bar{x}^{(k)})\|^2 \le \epsilon_{\text{UB}}$	$\Omega(\Delta L\sigma^2/(N\epsilon^2) + \Delta L/((1-\beta_{\text{H}})\epsilon))$ O5,A8,N8	$\widetilde{O}(\Delta L\sigma^2/(N\epsilon^2) + \Delta L/((1-\beta_{\text{H}})\epsilon))$ N8	COND	[8], Theorem 4	[8], Theorem 6
10	SUM PL SM SFO STAT	$\mathbb{E}[F(x_i) - F^\star] \le \epsilon$	$\Omega(K_{\text{loc}} + \min\{K_{\text{loc}}\kappa, \sqrt{K_{\text{loc}}/N\kappa_s}\}\log(1/\epsilon))$ A12,N9	$O((K_{\text{loc}} + \min\{K_{\text{loc}}\kappa, \sqrt{K_{\text{loc}}/N\kappa_s}\})\log(\Delta/\epsilon))$ N10	LOG	[9], Theorem 7; Table 4	[9], Corollary 2; Table 4
11	SUM NC SM SFO STAT	$\mathbb{E}\|\nabla F(x_i)\| \le \epsilon,\ \forall i$	$\Omega(K_{\text{loc}} + \min\{K_{\text{loc}}L_F, \sqrt{K_{\text{loc}}/N\bar{L}}\}\Delta/\epsilon^2)$ A28,N14	$O(K_{\text{loc}} + \min\{K_{\text{loc}}L_F, \sqrt{K_{\text{loc}}/N\bar{L}}\}\Delta/\epsilon^2)$ A28,N15	COND	[9], Theorem 3; Table 2; Remark 9	[9], Theorem 1; Corollary 1; Table 2
12	STO NC SM SFO STAT	$\mathbb{E}\|\nabla F(\hat{x})\| \le \epsilon$	$\Omega(\Delta L\sigma^2/(N\epsilon^4))$ A29,N16	$\widetilde{O}(\Delta L\sigma^2/(N\epsilon^4) + \Delta L/(\sqrt{\rho}\epsilon^2))$ N4,A2	GAP	[10], Corollary 1; Appendix Setting 1	[10], Theorem 3
13	STO PL SM SFO STAT S3	$\mathbb{E}[F(\hat{x}) - F^\star] \le \epsilon_{\text{LB}}$; $\frac{1}{T_{\text{out}}}\sum_{t=1}^{T_{\text{out}}}\mathbb{E}[F(\bar{x}^{(t)}) - F^\star] \le \epsilon_{\text{UB}}$ M5	$\Omega(\sigma^2/(\mu N\epsilon) + \sqrt{\kappa/\rho}\log(1/\epsilon))$	$\widetilde{O}(L\sigma^2/(\mu^2 N\epsilon) + \kappa/\sqrt{\rho}\log(1/\epsilon))$ A3	GAP	[6], Table 7	[6], Theorem 5
14	SUM C SM FO STAT S23	$F(\bar{x}) - F^\star \le \epsilon$ M8	$\Omega(\sqrt{LR^2/(\rho\epsilon)})$ N7	$\widetilde{O}(\sqrt{LR^2/(\rho\epsilon)})$ G17,N8,G18	COND	[11], Theorem 2 and paragraph following the theorem	[11], Theorem 1
15	SUM SC SM FO STAT	$\mathbb{E}V^t \le \epsilon$	EMPTY	$O(\max(L/\mu, \|\hat{W}\|/\omega(\hat{W}))\log(1/\epsilon))$ A13	LB?	/	[12], Appendix C, paragraph following Theorem C.1
16	SUM SC SM FO STAT S1	$\|x_i - x^\star\|^2 \le \epsilon$	EMPTY	$O(\log(1/\epsilon))$	LB?	/	[13], Theorem 1
17	SUM SC SM FO STAT	$\|x_i - x^\star\|^2 \le \epsilon$	EMPTY	$O(\log(1/\epsilon))$ A14	LB?	/	[14], Theorem 2; Theorem 1

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Table 1: Oracle-query complexity map. Tags describe the setting, Measure gives the target, LB/UB report oracle-query complexity only, Status marks exact matches or remaining gaps, and superscript markers link to Remark Notes. (Continued)

No.	Problem Type	Measure	Lower Bound	Upper Bound	Status	Reference (LB)	Reference (UB)
18	SUM SC SM FO TV S1,S15	$\|X - X^\star\|^2 \le \epsilon$	EMPTY	$O(\log(1/\epsilon))$ G9	LB?	/	[15], Theorem 1
19	SUM SC SM FO TV S24	$\|X - X^\star\|^2 \le \epsilon$	EMPTY	$O(\log(1/\epsilon))$ G15	LB?	/	[16], Theorem 4.1; Corollary 4.2
20	SUM SC SM FO TV	$\|X^T - X^\star\|_F^2 \le \epsilon$	EMPTY	$O(\kappa \log(R^2/\epsilon))$ O6	LB?	/	[17], Theorem 4.3
21	SUM SC SM FO TV	$\|\bar{x}_k - x^\star\|^2 \le \epsilon$	EMPTY	$O((L\tau_{\text{AC}}^2/\mu)\log(1/\epsilon))$	LB?	/	[18], Theorem 3.1; Corollary 3.1
22	SUM SC SM SFO STAT S22	$\mathbb{E}[F(x_i) - F^\star] \le \epsilon, \forall i$	EMPTY	$\widetilde{O}(\sigma^2/(N\mu\epsilon) + \sqrt{L/(\mu\rho_{\text{K}})}\log(\Delta_0/\epsilon))$ A24,A25,G13	LB?	/	[11], Theorem 5; Algorithm 4
23	SUM SC SM SFO STAT S1,S20	$\mathbb{E}\|\bar{x}_k - x^\star\|^2 \le \epsilon$	EMPTY	$O(\epsilon^{-1})$ G4,A19	LB?	/	[19], Theorem 2; Section IV-B
24	SUM SC SM SFO STAT	$\mathbb{E}\|z_{\text{VR}}^k - x^\star\|^2 \le \epsilon$	EMPTY	$O((\sqrt{K_{\text{loc}}\kappa_{s,\text{VR}}} + K_{\text{loc}})\log(1/\epsilon))$	LB?	/	[20], Corollary 5
25	SUM SC SM SFO STAT	$\mathbb{E}[\|\hat{x} - x^\star\|^2] \le \epsilon$	EMPTY	$O((K_{\text{loc}} + \kappa^2/\rho^2)\log(1/\epsilon))$ G10	LB?	/	[21], Theorems 1 and 2
26	SUM SC SM SFO TV	$\|X^T - X^\star\|^2 \le \epsilon$	EMPTY	$O((K_{\text{loc}} + \sqrt{K_{\text{loc}}\kappa_s})\log(1/\epsilon))$	LB?	/	[22], Corollary 4.2; Table 2
27	SUM SC NS FO STAT COMP PROX S18	$\mathbb{E}[\|X^t - X^\star\|_F^2] \le \epsilon$	EMPTY	$O(\max\{\kappa, p_{\text{comm}}^{-2}\}\log(\Psi^0/\epsilon))$ O1	LB?	/	[23], Theorem 1; Equation (11); following complexity paragraph; Table I
28	SUM SC NS FO STAT COMP PROX	$\|X^T - X^\star\|^2 \le \epsilon$	EMPTY	$O(\log(1/\epsilon))$ O1,G11	LB?	/	[24], Theorem 1
29	SUM SC NS FO TV COMP PROX S29	$\mathbb{E}\|\tilde{x}_i^T - x^\star\|^2 \le \epsilon, \ \forall i$ M9	EMPTY	$O(\log(1/\epsilon))$ O1,A30,N17	LB?	/	[25], Corollary 1; Equation (21)
30	SUM SC NS SUBG STAT	$F(\hat{x}) - F^\star \le \epsilon$	EMPTY	$O(M^2/(\mu\epsilon))$	LB?	/	[26], Table 3
31	SUM C SM FO STAT	$\|\sum_{i=1}^N F_i(\bar{x}_{i,T}) - \sum_{i=1}^N F_i(x^\star)\| \le \epsilon$	EMPTY	$O(C_{\mathbf{S}}C_0/(\alpha_{\text{DAGP}}\epsilon) + C_{\mathbf{S}}C_0C_2/(\eta_{\text{alg}}N\epsilon^2))$ O2	LB?	/	[27], Theorem 2
32	SUM C SM FO TV S17	$F(\bar{x}) - F^\star \le \epsilon$	EMPTY	$O(\sqrt{L/\epsilon})$ N2,N1	LB?	/	[28], Theorem 1; Corollary 3
33	SUM C SM SFO STAT S22	$\mathbb{E}[F(x_i) - F^\star] \le \epsilon, \ \forall i$	$\Omega(\sigma^2R^2/(N\epsilon^2))$ N18,N19	$\widetilde{O}(\sigma^2R^2/(N\epsilon^2) + \sqrt{LR^2/(\rho_{\text{K}}\epsilon)})$ A24,G1,G14,A25	GAP	[11], Paragraph following Theorem 2 *see also:* [51], Theorem 1	[11], Theorem 6
34	SUM C NS FO STAT COMP PROX	$F(\bar{y}_T) - F^\star \le \epsilon$	EMPTY	$O(\epsilon^{-1/2})$ O1,N2,N1	LB?	/	[29], Theorem 3; Corollary 4
35	SUM C NS SFO STAT	$\mathbb{E}[F(\hat{x}_i(T)) - F^\star] \le \epsilon$	EMPTY	$\widetilde{O}(M^2R^2/(\rho\epsilon^2))$	LB?	/	[30], Theorem 4
36	SUM C NS SFO TV COMP	$F(\tilde{x}_i^T) - F^\star \le \epsilon$	EMPTY	$\widetilde{O}(\epsilon^{-2})$ G2,O4,A15,G16	LB?	/	[31], Theorem 4
37	SUM C NS SFO TV COMP PROX	$\mathbb{E}[F(\tilde{x}^S) - F^\star] \le \epsilon$ M10	EMPTY	$\widetilde{O}(K_{\text{loc}} + (C_{\text{DPS}}/\epsilon)^{1/q_{\text{DPS}}})$ O1,A31,N20	LB?	/	[32], Theorems 2 and 3; Remark 1; Algorithm 1
38	SUM C NS SUBG STAT S26	$F(\hat{x}_i(T)) - F^\star \le \epsilon$	EMPTY	$\widetilde{O}(N^2M^2R^2/\epsilon^2)$	LB?	/	[30], Corollary 1
39	SUM C NS SUBG STAT S27	$F(\hat{x}_i(T)) - F^\star \le \epsilon$	EMPTY	$\widetilde{O}(M^2R^2/\epsilon^2)$	LB?	/	[30], Corollary 1
40	SUM C NS SUBG STAT S25	$F(\hat{x}_i(T)) - F^\star \le \epsilon$	EMPTY	$\widetilde{O}(NM^2R^2/\epsilon^2)$	LB?	/	[30], Corollary 1

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Table 1: Oracle-query complexity map. Tags describe the setting, Measure gives the target, LB/UB report oracle-query complexity only, Status marks exact matches or remaining gaps, and superscript markers link to Remark Notes. (Continued)

No.	Problem Type	Measure	Lower Bound	Upper Bound	Status	Reference (LB)	Reference (UB)
41	SUM C NS SUBG STAT S5	$\mathcal{E}_{\text{obj}}(\hat{y}) \le \epsilon$	EMPTY	$O(M^2\sqrt{N\log N}/\epsilon^2)$ A23	LB?	/	[33], Proposition 3.12
42	SUM C NS SUBG STAT	$\lvert F(\tilde{x}_i(t)) - F^\star\rvert \le \epsilon$	EMPTY	$\widetilde{O}(\epsilon^{-2})$	LB?	/	[34], Theorem 5; Corollary 2(b)
43	SUM C NS SUBG TV S6	$F(\hat{x}_j[T]) - F^\star \le \epsilon$	EMPTY	$O(1/\epsilon^2)$	LB?	/	[35], Theorem 2
44	SUM NC SM FO STAT S8	$\min_{0\le k<T} \lVert\nabla F(\bar{x}_u^k)\rVert \le \epsilon$	EMPTY	$O(1/(\eta^2\epsilon^2))$ A6,A7	LB?	/	[36], Theorem 1
45	SUM NC SM FO TV	$\frac{1}{T}\sum_{t=0}^{T-1}\lVert\nabla F(\bar{x}^{(t)})\rVert^2 \le \epsilon$	EMPTY	$O(\epsilon^{-1})$	LB?	/	[37], Theorem 2; Appendix A.9
46	SUM NC SM FO TV S9	$\min_{1\le s\le t}(N\lVert G_a(y^{s-1},\bar{z}^{s-1})\rVert_{\mathbb{E}}^2 + \lVert X^{s-1}-\mathbf{y}^{s-1}\rVert_{\mathbb{E}}^2) \le \epsilon$ M2	EMPTY	$O(C_{\text{DDA}}/\epsilon)$	LB?	/	[38], Theorem 2; Equation (23); Algorithm 1
47	SUM NC SM SFO STAT	$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\lVert\bar{g}(X^{(t)})\rVert_* \le \epsilon$ M2	EMPTY	$(p-1)T = O((p-1)\epsilon^{-(3p+1)/p})$ A1,A20	LB?	/	[39], Theorem 6; Equation (23)
48	SUM NC SM SFO TV	$\mathbb{E}\lVert\nabla F(\hat{x})\rVert^2 \le \epsilon^2$	EMPTY	$O(K_{\text{loc}} + \sqrt{K_{\text{loc}}}L_{\text{rms,loc}}\Delta/\epsilon^2)$ A27,N13	LB?	/	[22], Theorem 4.3; Corollary 4.5
49	SUM NC NS SFO STAT COMP PROX	$\mathbb{E}\lVert\mathbf{s}(\hat{x})\rVert^2 \le \epsilon^2$	EMPTY	$O((L\Delta/\epsilon^2)\max\{\sqrt{K_{\text{loc}}/N},1\} + \max\{K_{\text{loc}},\sqrt{NK_{\text{loc}}}\})$ O1,A17	LB?	/	[40], Theorem 3; Table 1
50	SUM PL SM FO STAT	$F(\bar{x}) - F^\star \le \epsilon$	EMPTY	$O(\theta^{-1}\log(c/(N\epsilon)))$	LB?	/	[41], Theorem 2; Equation (15)
51	SUM PL SM FO TV	$F(\bar{x}) - F^\star \le \epsilon$	EMPTY	$O(\kappa\log(\Delta/\epsilon))$ G6	LB?	/	[42], Theorem 3.1
52	STO SC SM SFO STAT S1,S19	$\mathbb{E}\lVert\bar{x}_{\pi_{\text{SAB}}}^k - x^\star\rVert^2 \le \epsilon$	EMPTY	$O(\epsilon^{-1})$ G4,A18	LB?	/	[43], Theorem 1 and following rate paragraph
53	STO SC SM SFO TV	$\sum_{t=0}^{T}\frac{w_t}{W_T}\mathbb{E}[F(\bar{x}^{(t)}) - F^\star] + \mu\mathbb{E}\lVert\bar{x}^{(T+1)} - x^\star\rVert^2 \le \epsilon$	EMPTY	$\widetilde{O}(\sigma^2/(\mu N\epsilon) + \sqrt{L}(\bar{\zeta} + \sigma\sqrt{\delta_{\text{net}}})/(\mu\delta_{\text{net}}\sqrt{\epsilon}) + L\log(1/\epsilon)/(\mu\delta_{\text{net}}))$ N5	LB?	/	[44], Theorem 2
54	STO SC NS SFO STAT	$F(\hat{x}) - F^\star \le \epsilon$	EMPTY	$O((M^2+\sigma^2)/(\mu\epsilon))$ G2	LB?	/	[26], Table 4
55	STO SC NS SFO TV S30	$\mathbb{E}[F(\bar{x}) - F^\star] \le \epsilon$	EMPTY	$O((M_\phi+\sigma)^2/(\mu\epsilon))$ A32,G19	LB?	/	[45], Theorem 2; Algorithm 1; Appendix C
56	STO C SM SFO TV	$(1/(T+1))\sum_{t=0}^{T}\mathbb{E}[F(\bar{x}^{(t)}) - F^\star] \le \epsilon$	EMPTY	$O(R^2(\sigma^2/(N\epsilon^2) + \sqrt{L}(\bar{\zeta} + \sigma\sqrt{\delta_{\text{net}}})/(\delta_{\text{net}}\epsilon^{3/2}) + L/(\delta_{\text{net}}\epsilon)))$ N5	LB?	/	[44], Theorem 2
57	STO C NS SFO STAT	$\Pr\{F(\hat{x}) - F^\star \le \epsilon\} \ge 1 - \delta_{\text{ht}}$	EMPTY	$O((M^2+\sigma^2)R^2/\epsilon^2)$ A33,G20,N21	LB?	/	[26], Table 4; high-probability discussion *see also:* [52], Section 4; Corollaries 3 and 4
58	STO C NS SFO TV	$\mathbb{E}[F(\bar{x}) - F^\star] \le \epsilon$	EMPTY	$O((M+\sigma)^2R^2/\epsilon^2)$	LB?	/	[45], Theorem 3
59	STO NC SM SFO STAT	$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\lVert\nabla F(\bar{x}^{(t)})\rVert^2 \le 3\epsilon^2 + \alpha_{\text{DSP}}\epsilon^3$	EMPTY	$O(\ell_{\text{DSP}}\sigma\epsilon^{-3} + \sigma^2\epsilon^{-2})$ A21	LB?	/	[46], Theorem 1; Corollary 1

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Table 1: Oracle-query complexity map. Tags describe the setting, Measure gives the target, LB/UB report oracle-query complexity only, Status marks exact matches or remaining gaps, and superscript markers link to Remark Notes. (Continued)

No.	Problem Type	Measure	Lower Bound	Upper Bound	Status	Reference (LB)	Reference (UB)
60	STO NC SM SFO STAT	$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|\nabla F(\bar{x}_{\pi_A}^t)\|^2 \le \epsilon$	$\Omega(\sigma^2 L\Delta/(N\epsilon^2))$ N22,N23	$O(\frac{\sigma^2 L\Delta}{N\epsilon^2} + \frac{(1+\log\kappa_A+\log N)L\Delta}{(1-\beta_A)\epsilon})$ G12	**GAP**	[10], Theorem 1; Appendix Setting 1	[47], Theorem 3
61	STO NC SM SFO TV S21	$(1/T)\sum_{t=0}^{T-1}\mathbb{E}[\|\nabla F(\bar{x}^{(t)})\|^2] \le \epsilon$	EMPTY	$O((1+\sigma^2)^2/(N\epsilon^2) + N\zeta_{\text{stat}}^2/\epsilon)$ G3,A22	**LB?**	/	[48], Theorem 2; Corollary 1
62	STO NC NS SFO STAT	$\mathbb{E}\|\nabla F(\hat{x})\|_{\delta_G} \le \epsilon$	EMPTY	$O(\rho^{-2}\delta_G^{-1}\epsilon^{-3})$	**LB?**	/	[49], Theorem 2; Theorem 1
63	SUM SC NS SFO STAT COMP PROX	$\mathbb{E}[F(\tilde{y}_{\text{PM}}^T) - F^\star] \le \epsilon$ M7	EMPTY	$O((K_{\text{loc}} + K_{\text{loc}}^{3/4}\sqrt{\kappa_{\text{snc}}})[1+\log(\Delta/\epsilon)])$ A26,O7,N12,O1	**LB?**	/	[50], Theorem 5.1; Corollary 5.3
64	SUM SC NS SFO TV	–	EMPTY	EMPTY	**EMPTY**	/	/
65	SUM C SM SFO TV	–	EMPTY	EMPTY	**EMPTY**	/	/
66	SUM C NS FO TV COMP PROX S28	$\mathbb{E}[F(\tilde{y}_{\text{DA}}^T) - F^\star] \le \epsilon$ M6	EMPTY	$O(\epsilon^{-1})$ N11,O1	**LB?**	/	[25], Corollary 2; Equation (22)
67	SUM NC NS FO STAT	–	EMPTY	EMPTY	**EMPTY**	/	/
68	SUM NC NS FO TV	–	EMPTY	EMPTY	**EMPTY**	/	/
69	SUM NC NS SFO TV	–	EMPTY	EMPTY	**EMPTY**	/	/
70	SUM NC NS SUBG STAT	–	EMPTY	EMPTY	**EMPTY**	/	/
71	SUM NC NS SUBG TV	–	EMPTY	EMPTY	**EMPTY**	/	/
72	SUM PL SM SFO TV	–	EMPTY	EMPTY	**EMPTY**	/	/
73	STO NC NS SFO TV	–	EMPTY	EMPTY	**EMPTY**	/	/
74	STO PL SM SFO TV	–	EMPTY	EMPTY	**EMPTY**	/	/

4 Remark Notes

Superscript markers next to main-table entries identify conditions that qualify a result's scope, assumptions, guarantee, target, or oracle-query accounting. **S**: Scope and topology; **O**: Oracle and side access; **A**: Additional assumptions; **G**: Guarantee regime; **M**: Metric and target qualification; **N**: Normalization and counting.

Scope and topology.

These notes identify restrictions on the communication graph, mixing model, topology family, or decentralized method class covered by the cited result.

S1 The cited result is stated for directed communication graphs rather than the unrestricted static or time-varying graph class.
S2 The cited lower bound uses the paper's sun-shaped time-varying graph construction.
S3 The cited result assumes the mixing parameter satisfies $\beta \leq \cos(\pi/N)$.
S4 The cited result assumes the mixing parameter satisfies $\beta \leq 1 - 1/N$.
S5 The cited result is restricted to two-dimensional grid or geometric network topologies.
S6 The packet-drop model starts from a fixed strongly connected directed graph and requires every original communication edge to succeed at least once in every B consecutive rounds.
S8 The cited result uses separate directed row-stochastic and column-stochastic mixing matrices.
S9 The cited result is formulated for a random-network model rather than an arbitrary deterministic time-varying network.
S11 The cited lower bound is proved for a linear-span method class on directed graphs.
S12 The cited lower bound is restricted to the paper's zero-initialized linear-span or local-memory decentralized first-order method class.
S13 Every time-varying mixing operator must preserve consensus and satisfy the paper's uniform contraction bound on the disagreement subspace.
S14 The cited result uses one fixed strongly connected directed graph with a primitive column-stochastic mixing matrix.
S15 The time-varying directed graphs are jointly strongly connected over every C-step window; the row- and column-stochastic weights have self-loops and uniform positive lower bounds.
S16 The cited result assumes a fixed connected undirected network and a symmetric positive-semidefinite gossip operator with the stated consensus kernel.
S17 The cited result assumes doubly stochastic mixing and a finite window parameter for which the paper's contraction factor $\sigma_\gamma < 1$.
S18 The cited multi-gossip result uses a symmetric doubly stochastic mixing matrix on a connected static graph.
S19 The cited S-AB result uses separate row- and column-stochastic directed matrices whose associated graphs have spanning trees with intersecting root sets.
S20 The cited S-ADDOPT result assumes a fixed strongly connected directed graph and requires each node to know its out-degree.
S21 The cited Push-SUM result assumes directed communication with strong connectivity over every fixed-length window.
S22 The cited result assumes a fixed symmetric doubly-stochastic gossip matrix and uses a reference-specific gap based on the second-largest eigenvalue, rather than the global one-step contraction gap.
S23 The cited lower bound uses an N-node path with $N \geq 6$, $\rho = \Theta(N^{-2})$, origin initialization, averaged output, and $d = \Theta(S\sqrt{\rho})$.
S24 The graph sequence is time-varying and directed; every union over B0 consecutive graphs is strongly connected, and each mixing matrix is column stochastic with the paper's out-degree weights.
S25 The cited upper bound is specialized to a two-dimensional grid topology.
S26 The cited upper bound is specialized to a cycle or path topology.
S27 The cited upper bound is specialized to an expander topology.
S28 The cited dual-averaging result uses independent-in-time random undirected doubly stochastic mixing with a uniform second-moment contraction; it is not an arbitrary deterministic time-varying-network guarantee.
S29 The source uses independent-in-time random undirected doubly stochastic matrices with $\lambda_{\max}(\mathbb{E}[P_t^\top P_t] - J) \leq \beta^2 < 1$, not arbitrary deterministic time-varying mixing.
S30 The source requires its per-time network contraction assumption, normalized here by χ_{TV}; joint connectivity alone is insufficient.

Oracle and side access.

These notes record oracle primitives or exact local operations required in addition to the local first-order calls counted in the displayed complexity.

O1 The cited result assumes exact access to a proximal operator in addition to the counted first-order oracle calls.
O2 The cited result requires exact local orthogonal projections onto the node constraint sets in addition to the counted gradients.
O3 The oracle model allows exact gradients of the local Fenchel conjugates in addition to ordinary local gradients; MSDA uses this conjugate-function gradient primitive.
O4 The cited result requires a mirror-map operation in addition to the counted stochastic first-order oracle calls.
O5 The cited lower bound restricts algorithms to the paper's partial-averaging communication model.
O6 The cited result relies on a finite consensus routine whose accuracy depends on the target accuracy.
O7 The cited PMGT-KatyushaX analysis is retained with a common initial point and an exactly consensual tracker equal to the global smooth-part gradient. This explicitly interprets the source's ambiguous composite-gradient initialization. Initialization costs K_{loc} component calls per node plus uncounted aggregation communication; ordinary unequal local-gradient initialization is not certified here.

Additional assumptions.

These notes list hypotheses that are stronger or more specific than the canonical row tags alone, including regularity, heterogeneity, initialization, and algorithmic restrictions.

A1 The cited result assumes order-p smoothness, which is stronger than ordinary first-order smoothness when $p > 1$.
A2 The cited upper bound assumes the paper's weak close-distribution condition, which bounds cross-node gradient dissimilarity only at the common initial point $x = 0$.
A3 The cited upper bound assumes a uniform bound on gradient dissimilarity across nodes.
A6 The cited result uses the paper's generalized-smoothness condition.
A7 The cited result assumes relative dissimilarity among local objectives.
A8 The cited lower bound is restricted to zero-respecting algorithms.
A9 The cited result requires $\nu = O(R/(\sqrt{N}\,\bar{\zeta}))$ to attain the displayed rate.
A10 The cited nonsmooth result uses local regularity: every local objective is convex and has its own finite Lipschitz constant.
A11 The cited nonsmooth lower bound is stated in the sufficiently high-dimensional regime required by its hard-instance construction.
A12 The cited lower bound holds only in its stated accuracy and smoothness-parameter regime, including the explicit separation between the mean and aggregate smoothness constants.
A13 The cited primal-dual linear-rate result assumes a fixed connected undirected graph, smooth strongly convex local functions, exact local gradients, and the prescribed initialization and stepsize conditions.
A14 The cited FROST result assumes a strongly connected static digraph, smooth strongly convex local objectives, and uncoordinated stepsizes in its admissible range.
A15 The cited Sub-Weibull result additionally assumes bounded local subgradients, Lipschitz regularizers, Sub-Weibull noise, strong graph connectivity, and convexity of the Bregman divergence in its second argument.
A17 The cited ProxGT-SR-E result assumes mean-squared smooth component gradients, unbiased sampling with bounded variance, consensual initialization, the stated spectral mixing condition, and exact prox.
A18 The cited S-AB asymptotic result assumes second-order growth at the optimum and differentiable stochastic gradients with the stated higher-moment and random-Lipschitz bounds.
A19 The cited S-ADDOPT result assumes conditionally unbiased stochastic gradients with uniformly bounded conditional variance.
A20 The cited DeMuon result assumes consensual initialization, bounded matrix noise, a primitive doubly stochastic mixer, coupled extrapolation and momentum parameters, horizon-dependent stepsizes, and the theorem's minimum iteration threshold.
A21 The cited D-SPIDER-SFO result assumes smooth local objectives, unbiased bounded-variance stochastic gradients, the stated spectral mixing condition, consensual initialization, and the prescribed batch and stepsize rules.
A22 The cited Push-SUM result assumes bounded stochastic-gradient variance and a uniform global bound on cross-node gradient dissimilarity.
A23 The optimization consequence treats its initial-distance factor as bounded and requires every node to know a common size estimate U satisfying $U \geq N$ and $U = O(N)$.
A24 The objective is a deterministic node sum; stochasticity enters only through the local oracle.
A25 Heterogeneity is bounded only at the optimum, not uniformly for all x.
A26 The cited sum-of-nonconvex result assumes a strongly convex total composite objective, a convex smooth aggregate, and component curvature bounds defining κ_{snc}. Individual components need not be convex. The audited subcase requires a strictly positive strong-convexity modulus for the smooth aggregate because the proof divides by that modulus.
A27 The cited GT-PAGE analysis assumes the component mean-squared smoothness inequality separately at every node, with bound $L_{\mathrm{rms,loc}}$; a network-averaged component bound alone is insufficient. The display follows the network-average objective used in the nonconvex analysis, despite the source introduction's sum notation.
A28 Here L_F is the smoothness of the global objective F, while $\bar{L}$ is component mean-squared smoothness. The cited DEAREST result also assumes a connected undirected static network with a symmetric doubly stochastic positive-semidefinite mixing matrix and common initialization.
A29 The retained Lu–De Sa component uses the paper's zero-respecting/linear-span class, fresh independent unbiased variance-bounded stochastic gradients, $\sigma > 0$, and the stated sufficiently small-ϵ regime.

A30 Row 29 assumes a shared closed convex proximable h, exact local gradients and prox steps, common initialization, and a fixed $a \in (0, 1/\mu)$ satisfying source restrictions (16) and (18). The exact prefactor is $C_{29} = (2/a)(2C/\mu + D)$.
A31 The cited DPSVRG theorem assumes componentwise smooth finite sums plus a shared proximable h, exact prox, time-varying b-connected mixing, bounded estimator/local-gradient/subgradient errors, and source parameters independent of ϵ. Its full error constant is $C_{\mathrm{DPS}} = (1 + r_{\mathrm{DPS}}/(2n_{0,\mathrm{DPS}}))\Delta + \|x^0 - x^\star\|^2/[n_{0,\mathrm{DPS}}(2\alpha_{\mathrm{DPS}} - 8\alpha_{\mathrm{DPS}}^2 L)] + C_{\mathrm{err,DPS}}/[n_{0,\mathrm{DPS}}(r_{\mathrm{DPS}}\beta_{\mathrm{DPS}} - 1)]$.
A32 For row 55, $F_i(x) = \phi_i(x) + \mu\|x\|^2/2$. The bound M_ϕ applies only to subgradients of convex ϕ_i; its oracle is conditionally unbiased with variance at most σ^2, while the quadratic term is known and evaluated exactly.
A33 Row 57 uses a common compact convex Euclidean domain of diameter at most R, exact projections, convex M-Lipschitz local objectives, and conditionally unbiased worker-independent noise satisfying $\mathbb{E}[\exp(\|g_i - v_i\|^2/\sigma^2) \mid \mathcal{F}] \le e$.

Guarantee regime.

These notes explain the probabilistic, initialization, stepsize, finite-time, or asymptotic regime in which the displayed guarantee holds.

G1 The cited result provides the displayed guarantee only from a prescribed warm start.
G2 The displayed complexity is a high-probability guarantee rather than an expectation guarantee.
G3 The displayed rate is the sufficiently-large-T simplification in Corollary 1; Theorem 2 contains additional finite-time initial-error, network, variance, and heterogeneity terms.
G4 The cited result uses a diminishing stepsize schedule.
G6 The displayed rate assumes an exact oracle (zero oracle bias); the consensus tolerance is chosen with the target accuracy so the additive error floor is at most the target, affecting communication rounds but not the displayed per-node gradient-query order.
G7 The cited directed lower bound requires β_π to be bounded away from zero, $\beta_\pi \le 1 - 1/N$, and $\log \kappa_\pi = \Omega(N(1 - \beta_\pi))$; it is not uniform over all parameter tuples.
G9 The displayed logarithmic iteration complexity suppresses a contraction factor whose constants depend on graph connectivity, mixing weights, objective regularity, and the admissible stepsize.
G10 The cited GT-SVRG result requires the prescribed stepsize, inner-loop length, and initialization conditions.
G11 The displayed ϵ-only logarithmic order treats the cited result's contraction factor as fixed; otherwise the iteration count is proportional to $\log(1/\epsilon)/\log(1/\gamma)$.
G12 The cited MG-PullDiag-GT result requires the specified stepsize, minibatch, inner-loop, and directed-mixing parameter choices.
G13 The first restart epoch starts in consensus; later epochs use warm-started outputs.
G14 The cited warm-start result requires its two explicit RMS disagreement inequalities.
G15 Each node uses a positive time-invariant but node-dependent stepsize. The stepsize maximum and ratio must satisfy Theorem 4.1 or Corollary 4.2; the displayed logarithmic oracle order treats the induced contraction factor $\lambda \in (0, 1)$ as fixed, and otherwise carries a factor $1/\log(1/\lambda)$.
G16 The displayed Sub-Weibull complexity suppresses confidence- and tail-dependent polylogarithmic factors, including powers of $\log(T/\delta_{\mathrm{ht}})$ governed by the tail index, where δ_{ht} is the failure probability.
G17 The row-14 upper bound specializes the cited stochastic theorem to $\sigma = 0$, uses its integer budget schedule, and hides the theorem's logarithmic factor.
G18 The matching order is conditional on the source nonblocking coupled clock and on the path/linear-span lower-bound class; it is not an unrestricted oracle-query LOG claim.
G19 The full achievable expression contains $1 + \chi_{\mathrm{TV}} M_\phi/\sqrt{\mu\epsilon} + (M_\phi + \sigma)^2/(\mu\epsilon)$. The main cell records its fixed-network, sufficiently small-ϵ leading oracle order, with integer-rounding constants suppressed.
G20 The displayed row-57 order fixes the graph and failure probability and records the leading small-ϵ term. A conservative full form is $\widetilde{O}(1 + (M^2 + \sigma^2)R^2/\epsilon^2 + MR\sqrt{\chi_L}/\epsilon)$, with confidence logarithms suppressed.

Metric and target qualification.

These notes clarify how the cited result's controlled metric or merit target differs from, or is related to, the measure displayed in the table.

M1 The cited result controls a Bregman merit function rather than the ordinary objective gap or Euclidean distance.
M2 Only the stationarity component of the cited result is retained in the displayed metric; its other residual terms are omitted.
M3 The displayed metric follows from the equivalence between the strongly convex smooth metric in the source and the table measure.
M4 The cited lower bound uses a node-local output selected from local memory, whereas the cited upper bound uses the global average output; only the oracle-order expressions are compared.
M5 The cited lower bound permits an arbitrary algorithm output, whereas the cited upper bound controls the iteration-average metric over network-average iterates; only the resulting oracle orders are compared.
M6 The cited convex dual-averaging result controls the time average of auxiliary prox points formed from averaged dual accumulators, not the local returned iterates or their arithmetic average. Forming this virtual output requires aggregation and prox side work beyond the counted local gradients.
M7 The cited PMGT-KatyushaX guarantee concerns the objective gap at the network-average primal output, not each individual node's output.
M8 The source upper bound controls every node's objective gap; convexity then gives the displayed bound for $F(\bar{x})$.
M9 The guarantee is for every node's weighted output $\tilde{x}_i^T = A_T^{-1} \sum_{t=1}^{T} a_t x_i^t$, with $a_t = a(1 - a\mu)^{-t}$, rather than an unspecified final iterate.
M10 The source controls the virtual centralized epoch output $\tilde{x}^S$, not every node's returned local iterate.

Normalization and counting.

These notes explain how iteration, batch, and sample counts in cited results are converted into per-node oracle queries and which communication work is excluded.

N1 The displayed complexity counts local gradient evaluations and does not count the additional consensus rounds.
N2 Each outer iteration performs multiple gossip or consensus rounds, which are not counted as oracle queries in the displayed rate.
N3 The displayed count $N_{\mathrm{SFO}} = T_{\mathrm{out}} R$ includes the R samples per outer iteration; each outer iteration also performs two R-round gossip phases, counted separately as communication.
N4 The displayed stochastic-oracle complexity is normalized to local batch size $B = 1$.
N5 The displayed effective per-iteration network rate normalizes the cited result's block-contraction parameter by its mixing-window length.
N7 The displayed per-node epsilon form is derived from the communication-round bound in its nonconstant-error branch.
N8 The displayed network-dependent lower-bound term is an oracle projection of a coupled round model in which each node performs one local query and one gossip step per round; it is not an unconditional communication-free oracle lower bound.
N9 The displayed lower bound divides an aggregate LIFO lower bound by N; it applies to average and worst-node workload, not every node individually. The full computation-round lower bound additionally contains $L\Delta/\epsilon^2$ in the nonconvex row and $\kappa \log(1/\epsilon)$ in the PL row. These serial-depth terms are excluded from the oracle count because partial participation permits different nodes to carry different rounds.
N10 The displayed upper bound divides expected aggregate LIFO calls by the node count under uniform node-component sampling; it is an expected per-node workload bound rather than an expected-maximum workload bound.
N11 The cited dual-averaging algorithm uses one initial local gradient and at most one fresh local gradient per iteration. The displayed epsilon-only order holds for a fixed admissible stepsize, network, smoothness and initialization; their full coefficient is retained in the evidence ledger.
N12 The cited PMGT-KatyushaX rate counts expected component-gradient evaluations per node, including full-gradient snapshots, both evaluations in sampled gradient differences and initialization. The geometric inner-loop length and batch size of order $\sqrt{K_{\mathrm{loc}}}$ give an expected epoch cost of order K_{loc}; multi-consensus and exact prox work are separate.
N13 The cited GT-PAGE count is expected work at each node, including the initial full local gradient and stochastic refresh/difference calls; it is not an expected maximum over nodes or a deterministic stopping budget.
N14 The row-11 lower bound is for the cited IFO/LIFO span class in its small-ϵ regime. Dividing aggregate LIFO work by N proves an average or some worst-node requirement, not the same lower bound at every node; the separate computation-round term is excluded.
N15 Uniform node-component sampling makes the displayed DEAREST+ order the expected local query count at every node; communication rounds are counted separately.

N16 With $B = 1$, only the Setting-1 sample/noise component is normalized from total work to a uniform-cap or worst-node per-node query count. Corollary 1's spectral branch counts communication rounds and is not promoted to N_{SFO}.
N17 The $O(\log(1/\epsilon))$ display fixes a, μ, C_{29} and the network constants and includes initialization; it does not assert an explicit unrestricted graph dependence.
N18 Only the classical stochastic-oracle term is promoted. With total sample budget $N_{\mathrm{tot}} = NQ$, the minimax error $\Omega(\sigma R/\sqrt{N_{\mathrm{tot}}})$ implies $Q = \Omega(\sigma^2 R^2/(N\epsilon^2))$. It remains valid with free gossip and gives a uniform-cap or worst-node requirement, not the same burden at every node; it is nontrivial only for $\sigma > 0$.
N19 The path/network lower-bound term is not entered because converting its communication rounds to oracle calls uses the source's one-query/one-gossip coupled clock and deterministic path linear-span class.
N20 The source count is $K_{\mathrm{loc}}S + 2\sum_{s=1}^{S}\lceil n_{0,\mathrm{DPS}}\beta_{\mathrm{DPS}}^{s}\rceil$; the display hides epoch logarithms and excludes multi-consensus and exact-prox work.
N21 One stacked stochastic-oracle evaluation in the cited decentralized method means one query at each node; the per-worker count is therefore not divided by N.
N22 The Lu–De Sa Setting-1 hard instance yields the squared-stationarity noise term after total-query/N normalization under a uniform cap or worst-node count. It permits free communication but assumes fresh unbiased bounded-variance queries, a zero-respecting/linear-span class, $\sigma > 0$, and sufficiently small ϵ.
N23 The separate β_A, κ_A term from the row-60 reference is not promoted: its argument counts a coupled communication/iteration clock rather than unrestricted per-node stochastic-oracle queries.

5 Summary

Across 74 rows, this table presents 47 of the 96 vanilla settings and omits the remaining 49 empty settings. Evidence covers 37 of the 47 presented settings (37 of 96 in the full grid), while 10 presented settings are explicitly **EMPTY**. The table retains 16 lower-bound and 64 upper-bound claim sides (80 total), citing 52 distinct arXiv papers; 12 papers are cited on lower-bound sides and 51 on upper-bound sides, including supporting provenance and with overlap between the two paper sets. Row statuses are **EXACT**: 2; **COND**: 9; **LOG**: 1; **GAP**: 4; **UB?**: 0; **LB?**: 48; **EMPTY**: 10; status totals count rows and paper totals are deduplicated by arXiv ID.

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